

KRISH AI - Excel Knowledge Base

Plain selectable text knowledge base for Krish AI. Designed for PyMuPDF extraction and RAG chunking.

Excel Fundamentals

Excel is a spreadsheet application used for data entry, calculation, analysis, visualization, and reporting.

A workbook contains worksheets. A worksheet contains rows and columns. A cell is the intersection of a row and column. A range is a group of cells. A formula begins with an equals sign.

Cell References

=A1+B1

=A1*\$B\$1

Relative reference A1 changes when copied. Absolute reference \$A\$1 stays fixed. Mixed references such as A\$1 and \$A1 fix one dimension.

Basic Operators

=A1+B1

=A1-B1

=A1*B1

=A1/B1

=A1^2

Excel uses +, -, *, /, and ^ for arithmetic.

Basic Functions

=SUM(B2:B20)

=AVERAGE(B2:B20)

=MIN(B2:B20)

=MAX(B2:B20)

=COUNT(B2:B20)

SUM adds numbers. AVERAGE calculates the mean. MIN and MAX return extremes. COUNT counts numeric cells.

COUNTA and COUNTBLANK

=COUNTA(A2:A20)

=COUNTBLANK(A2:A20)

COUNTA counts non-empty cells. COUNTBLANK counts blank cells.

IF and Logical Functions

=IF(B2>=50, "Pass", "Fail")

=AND(B2>=50, C2>=50)

=OR(B2>=50, C2>=50)

=NOT(B2>=50)

IF returns different results based on a condition. AND requires all conditions. OR requires at least one. NOT reverses a condition.

Conditional Aggregation

=SUMIF(A2:A20, "Sales", B2:B20)

=SUMIFS(C2:C20, A2:A20, "Sales", B2:B20, ">50000")

=COUNTIF(A2:A20, "Sales")

=COUNTIFS(A2:A20, "Sales", B2:B20, ">50000")

SUMIF and COUNTIF use one condition. SUMIFS and COUNTIFS support multiple conditions.

AVERAGEIF and AVERAGEIFS

=AVERAGEIF(A2:A20, "Sales", B2:B20)

=AVERAGEIFS(C2:C20, A2:A20, "Sales", B2:B20, ">50000")

These functions calculate conditional averages.

Text Functions

=LEFT(A2, 3)

=RIGHT(A2, 4)

=MID(A2, 2, 5)

=LEN(A2)

=UPPER(A2)

=LOWER(A2)

=TRIM(A2)

LEFT, RIGHT, and MID extract text. LEN counts characters. UPPER and LOWER change case. TRIM removes extra spaces.

Text Combining and Searching

=A2&" "&B2

=CONCAT(A2, " ", B2)

=TEXTJOIN(" ", TRUE, A2:A5)

=FIND("@", A2)

=SEARCH("sales", A2)

FIND is case-sensitive. SEARCH is not case-sensitive. TEXTJOIN combines values using a delimiter.

Date and Time

=TODAY()

=NOW()

=YEAR(A2)

=MONTH(A2)

=DAY(A2)

Excel stores dates as serial numbers. TODAY returns the current date. NOW returns date and time. YEAR, MONTH, and DAY extract components.

Lookup Functions

=VLOOKUP(E2, A2:C20, 3, FALSE)

=XLOOKUP(E2, A2:A20, C2:C20, "Not Found")

=INDEX(C2:C20, MATCH(E2, A2:A20, 0))

VLOOKUP searches vertically. XLOOKUP is a flexible modern lookup function. INDEX and MATCH can be combined for flexible lookups.

IFERROR

```
=IFERROR(XLOOKUP(E2, A2:A20, B2:B20), "Not Found")
```

IFERROR returns an alternative result when a formula produces an error.

Sorting and Filtering

Sorting rearranges records. Filtering displays only records that meet selected conditions. Filters are useful for exploring large datasets.

Excel Tables

An Excel Table is a structured data range with headers, filters, automatic expansion, and structured references.

```
=SUM(SalesTable[Revenue])
```

Good raw data has one header row, one record per row, and one variable per column. Avoid merged cells inside raw datasets.

Data Validation

Data Validation controls what users can enter. It can restrict values to lists, numbers, dates, or custom rules. Drop-down lists improve consistency.

Conditional Formatting

Conditional Formatting applies visual rules to cells. Common uses include duplicate highlighting, thresholds, data bars, color scales, and status indicators.

Data Cleaning

Data cleaning includes removing unwanted spaces, standardizing text, correcting inconsistent categories, handling blanks, checking data types, and identifying duplicates.

```
=TRIM(A2)
```

```
=CLEAN(A2)
```

Remove Duplicates can remove repeated records. Keep a copy of original data before destructive cleaning.

Flash Fill

Flash Fill recognizes patterns from examples and fills remaining values. It can help split names, combine text, extract codes, and standardize patterns.

PivotTables

A PivotTable summarizes data without requiring many formulas. Fields can be placed in Rows, Columns, Values, and Filters.

Typical analysis includes sales by region, order count by category, average revenue by month, and profit by product.

PivotCharts and Slicers

A PivotChart visualizes PivotTable results. A slicer provides clickable filtering. A timeline can filter date fields in compatible PivotTable reports.

Charts

Column and bar charts compare categories. Line charts show trends over time. Scatter charts show relationships between numeric variables. Pie charts show parts of a whole but should be used carefully when there are many categories.

Excel Errors

#DIV/0! occurs when a formula divides by zero. #N/A often means a lookup found no match. #VALUE! indicates an incompatible value or data type. #REF! indicates an invalid reference. #NAME? often indicates an unrecognized name or function.

Use IFERROR when a controlled display is needed, but do not hide errors blindly because they may indicate data-quality problems.

Dynamic Array Functions

```
=FILTER(A2:C20, C2:C20="Sales")
```

```
=SORT(A2:C20, 2, 1)
```

```
=UNIQUE(A2:A20)
```

FILTER returns matching rows. SORT sorts an array. UNIQUE returns distinct values. Availability depends on Excel version.

Power Query

Power Query is Excel's data import and transformation tool. It can connect to files and databases and perform repeatable transformations.

Common transformations include changing data types, filtering rows, removing columns, splitting columns, merging queries, appending queries, grouping data, replacing values, and removing duplicates.

Data Analysis Workflow

A practical workflow is: import data, inspect columns and data types, clean inconsistent values, handle missing data, remove duplicates when appropriate, create calculated fields, summarize with formulas or PivotTables, visualize important patterns, validate results, and prepare the report.

Excel Best Practices

Keep raw data separate from analysis. Use clear column names. Avoid merged cells in datasets. Keep one record per row. Use Excel Tables. Use consistent date and number formats. Keep formulas understandable. Validate important totals against source data.